

Book Topic Teaching Activity

Purpose

Choose one topic represented in the course weekly schedule. Study the relevant textbook chapter and pages until you can explain the concepts in your own words. Then create a small Python-based activity that lets classmates actively understand and practice those concepts.

This is more than a book report or a code demo: you are designing a short learning experience. Your goal is for peers to be able to follow your instructions, run or use your activity, and leave with a clearer grasp of the topic. Make it creative and inviting, while keeping the Python learning goal clear.

Prepare

Use the course textbook that corresponds to your chosen scheduled topic. Your activity and short introduction should:

- Identify the schedule topic, textbook chapter, and printed pages you studied.
- State the concepts classmates will practice and why those concepts are useful in Python.
- Teach two or three key ideas accurately in your own words.
- Give peers a hands-on Python task, puzzle, mini-project, simulation, guided code exercise, or other practice activity tied directly to the topic.
- Include clear steps, any starter files or sample data needed, expected result/checkpoint, and a way for peers to check their understanding.
- Point out one common mistake or misconception and how to avoid it.
- Connect the topic to an in-class activity, weekly homework, or possible real application.
- Credit the textbook and any other sources or AI assistance.
- Close with a brief reflection or check-for-understanding prompt.

Keep the activity achievable in the time the instructor provides. Test the activity yourself from a clean start. Then ask at least one peer to follow the README without verbal help; record where they get stuck and improve the directions. If you run code live, test it beforehand and have saved output or screenshots as a backup. Slides can support the activity, but should not replace peer practice.

Git Repository and Submission

Create a Git repository for your teaching activity and commit meaningful progress as you work. Submit the repository link through Canvas using the course access directions. Your repository should include:

- A README with the topic, learning goal, textbook chapter/pages, prerequisites, setup/run instructions, activity steps, expected outputs or checkpoints, and how peers can check their work.
- The Python source files and any small sample data required to run the activity.

- A `requirements.txt` if you use third-party packages; otherwise state the Python version and run command.
- A short AI-use/source note identifying relevant outside materials and how you verified the activity.

Use a private repository unless the instructor has approved public sharing. Do not commit credentials, API keys, or private information. Make sure classmates and the instructor can access the repository during the activity.

AI Use

You may use AI to brainstorm an explanation, create practice questions, or review your slides. Verify technical claims and code against the textbook and by running the example. Keep a brief note of meaningful AI assistance and disclose it at the end. Do not paste private information or submit an AI-written presentation you cannot explain.

In-Class Teaching Format

- Briefly introduce the need/topic and the textbook material you used.
- Lead classmates through your activity; allow time for them to try the Python task themselves.
- Explain the concept and common pitfall as they work, then debrief the result.
- The instructor will announce the time limit and presentation date.
- Submit the repository link and any supporting slides through Canvas by the announced deadline.

Rubric (100 points)

Area	Points	Full-credit evidence
Topic mastery and textbook connection	20	Concepts are accurate, explained in the student's own words, and connected to the chosen schedule topic and cited book pages.
Learning activity and Python practice	30	Peers actively practice the target concepts through a purposeful, creative, appropriately scoped activity that works.
Peer usability and learning checks	20	Instructions are independently followable; prerequisites and expected results are clear; the activity checks whether peers understood.

Area	Points	Full-credit evidence
Professional Git repository	15	Repository is accessible as directed, organized, and contains working code, a complete README, setup details, and meaningful commit history.
Facilitation and engagement	10	Student introduces, guides, and debriefs the experience clearly; peers have time to participate.
Sources and AI transparency	5	Textbook pages and outside materials are credited; meaningful AI assistance is disclosed and verified.

Before You Present

- Test the repository from a clean start and ask a peer to follow the README.
- Verify the code, data, expected results, and textbook page references.
- Check that classmates can access the repository and understand what to do.
- Be ready to explain why your activity helps peers learn this concept and what you changed after testing it.